

Nirma University

Institute of Technology

Semester End Examination (IR), January - 2022

Master of Computer Application, Semester - I

3MCA102 Data Structures

Roll /
Exam No.

Supervisor's initial
with date

Time: 2 Hours

Max. Marks: 50

Instructions:

1. Attempt all questions.
2. Figures to right indicate full marks.
3. Draw neat sketches wherever necessary.
4. Assume suitable data wherever necessary and mention the same.

Q-1.

Answer the following.

[18]

- (A) CO-1,2,4
BL-3 Given an array `arr[1.....10][1.....15]` with base value **200** and the size of each element is **2 Byte** in memory find the address of `arr[8][9]` with the help of column-major order. What would be the address of `arr[8][9]` with the help of row major order also. (6)
- (B) CO-1,2,4
BL-4 Consider the following arithmetic expressions M, written in postfix notation: (6)
- M: $8\ 2\ 3\ ^\wedge\ /\ 2\ 3\ *\ +\ 5\ 1\ *\ -$
- Evaluate the above expression showing stack status after every step.
- (C) CO-1,2,4
BL-3 How to implement priority queue using 2D array? Write an algorithm for the same. (6)
- OR**
- (C) CO-1,2,4
BL-3 Develop an algorithm to add an element at the end of doubly link list. (6)

Q-2.

Answer the following.

[14]

- (A) CO-1,2,3,4
BL-3 Perform step by step tracing to sort the following data using quick sort. (7)
- 30, 26, 37, 58, 13, 10, 21
- (B) CO-1,2,3,4
BL-4 Apply Linear Search and Binary Search algorithm to Search element 51 for the following input array. (7)
- Array : 5 10 12 15 18 24 27 29 31 36 37 45 49 51 58

Also Justify which searching method will take less time for the above array to search element 51.

Q-3.

Answer the following.

[18]

- (A) **CO-1,2,4** Draw the B Tree of order 3 that results from inserting the (9)
BL- 5 following sequence into an initially empty tree.
 52, 20, 36, 1, 68, 75, 43, 95, 85, 42, 16, 19, 23, 81, 10
 Show all the intermediate steps.
- (B) **CO-1,2,4** Construct the binary tree from the given sequences. (9)
BL- 3
- (i) Pre-order = {10, 5, 2, 6, 14, 12, 15}
 - In-order = {2, 5, 6, 10, 12, 14, 15}
 - (ii) In-order = {D, B, E, A, F, C, G}
 - Post-order = {D, E, B, F, G, C, A}

OR

- (B) **CO-1,2,4** Apply Prim's and Kruskal's Algorithm for the following graph. (9)
BL- 3 Write the cost of Minimum spanning tree in each case.

